

PRD: SINicare

A GenAI Daily Digital Companion for Senior Citizens

Status: Draft v1 (MVP) | **Platform:** Responsive web app (desktop, tablet, mobile) | **Languages:** English, Hindi

1. Summary

SINicare is a voice-first web app where seniors say or type what they need, and get calm, simple, step-by-step help with everyday digital tasks: understanding messages and documents, spotting scams, filling forms, navigating websites, and remembering tasks. Family members can optionally be added as trusted contacts.

2. Problem

Most digital products assume tech fluency. For seniors this means:

- **Confusion:** dense forms, jargon-heavy documents, cluttered websites.
- **Risk:** phishing SMS, fake bank/KYC alerts, and impersonation scams disproportionately target older adults.
- **Dependence:** simple tasks need a child or neighbour, which erodes confidence and independence.

3. Target Users

Persona	Description	Key need
Primary: Senior user (60+)	Owns a smartphone or shared family device; low-to-moderate digital confidence; may prefer Hindi; may have reduced vision, hearing, or dexterity	Plain-language help without feeling judged
Secondary: Trusted contact	Adult child, relative, or caregiver, often living elsewhere	Peace of mind without taking over

4. Goals and Non-Goals

Goals (MVP)

1. A senior completes a first useful task within 2 minutes of opening the app, with no tutorial.
2. Suspicious messages are flagged clearly and correctly, and the senior is told what to do next.
3. Seniors finish everyday tasks (understand, fill, remind) without asking another person.

Non-goals (MVP)

- Autonomous actions (payments, sending messages, submitting forms on the user's behalf).
- Medical, legal, or financial advice. SINicare explains and signposts; it does not advise.
- Native apps, additional languages beyond English and Hindi, integrations with banks or government portals.

5. Product Principles

Simple • Trustworthy • Safe • Accessible • Independent • Human-centered

Design rules that follow: one primary action per screen; never guess silently (say "I'm not sure"); the senior always stays in control; the AI never pressures, rushes, or shames.

6. MVP Features

#	Feature	What it does	Pri.	Acceptance criteria (summary)
F1	AI conversational companion	Home screen with one large "Ask" box and mic button; answers in short, plain language with suggested follow-ups	P0	Replies at roughly grade-6 reading level; no jargon without explanation; responds in the language the user used

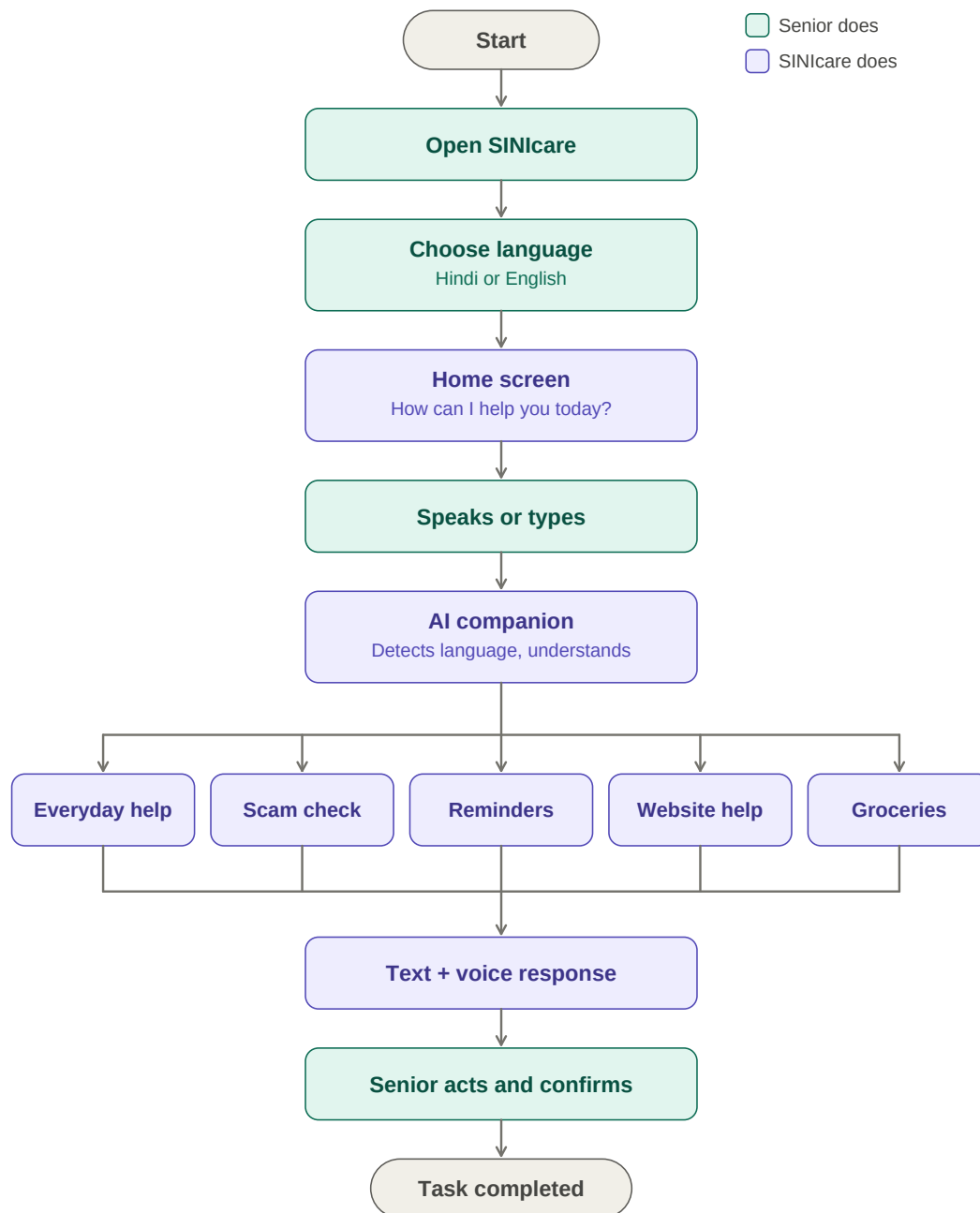
#	Feature	What it does	Pri.	Acceptance criteria (summary)
F2	Voice input + text-to-speech	Speak to ask; every answer can be read aloud (auto-read is a setting)	P0	Works in English and Hindi; adjustable speech speed; clear fallback to typing if mic is denied
F3	Message and document simplification	Paste text, upload a photo/PDF, or share a screenshot; get a 3-part output: <i>What this is</i> / <i>What it wants from you</i> / <i>What to do next</i>	P0	Key dates, amounts, and deadlines are called out; uncertainty is stated explicitly
F4	Scam and suspicious-message check	Traffic-light verdict (Looks safe / Be careful / Likely scam) with reasons in plain words and a "what to do now" action	P0	Never says "safe" with false certainty; always advises not to share OTP, PIN, or passwords; offers "Ask my family"
F5	Step-by-step form and task help	Converts a form or task into one question at a time; explains each field; shows a final review checklist	P0	One step per screen; back/undo always available; app never submits anything for the user
F6	Guided website navigation	User describes a goal ("pay electricity bill"); SINicare gives numbered steps with what to look for on screen. Optional screenshot upload for "where do I click?"	P1	Steps reference visible labels; warns when a page looks unofficial
F7	Reminders and daily tasks	Create by voice or text ("remind me to take a call at 5"); simple Today list with big Done buttons	P0	Browser notification plus optional email/SMS; snooze; read-aloud of today's list
F8	Multilingual (EN + HI)	Full UI and AI in English and Hindi; switch at any time; supports Hinglish input	P0	All UI strings localized; language toggle on every screen
F9	Trusted-contact assistance	Senior invites a contact; can tap "Ask my family" to share a message, scam verdict, or question. Contact can send reminders	P1	Consent-based and revocable; senior sees exactly what is shared; contact never sees chat history by default

7. User Flows

The flows below show how a senior moves through SINicare and where the safety and control points sit. Teal boxes are the senior's steps, purple boxes are SINicare's; a thick border marks a decision that only the senior can make.

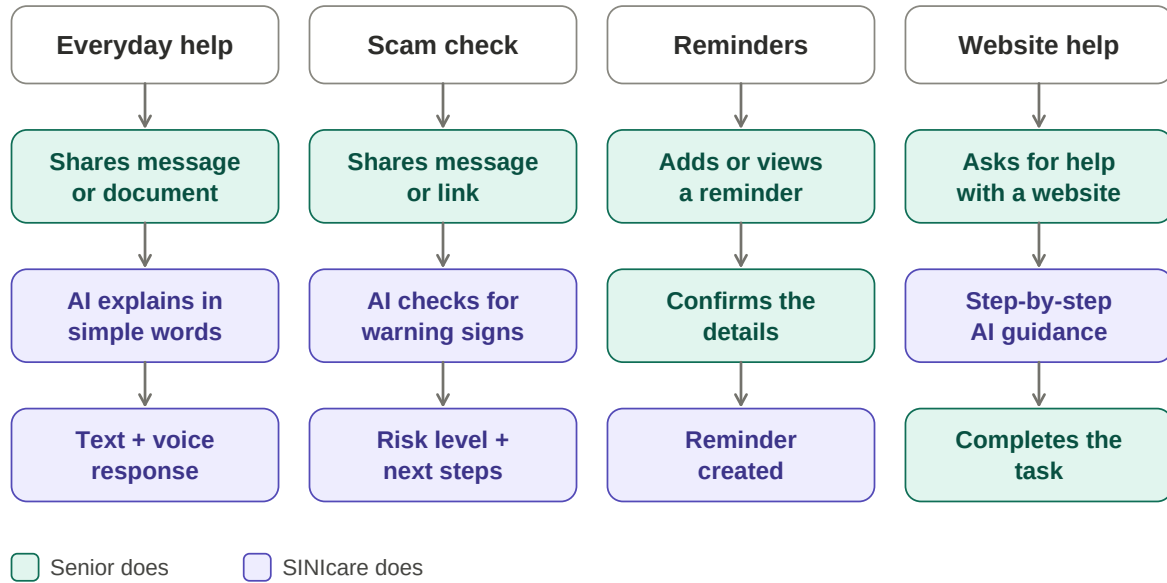
7.1 Main journey

From opening the app to a completed task. Voice or text input, language detection, and the text + voice response are common to every help area.



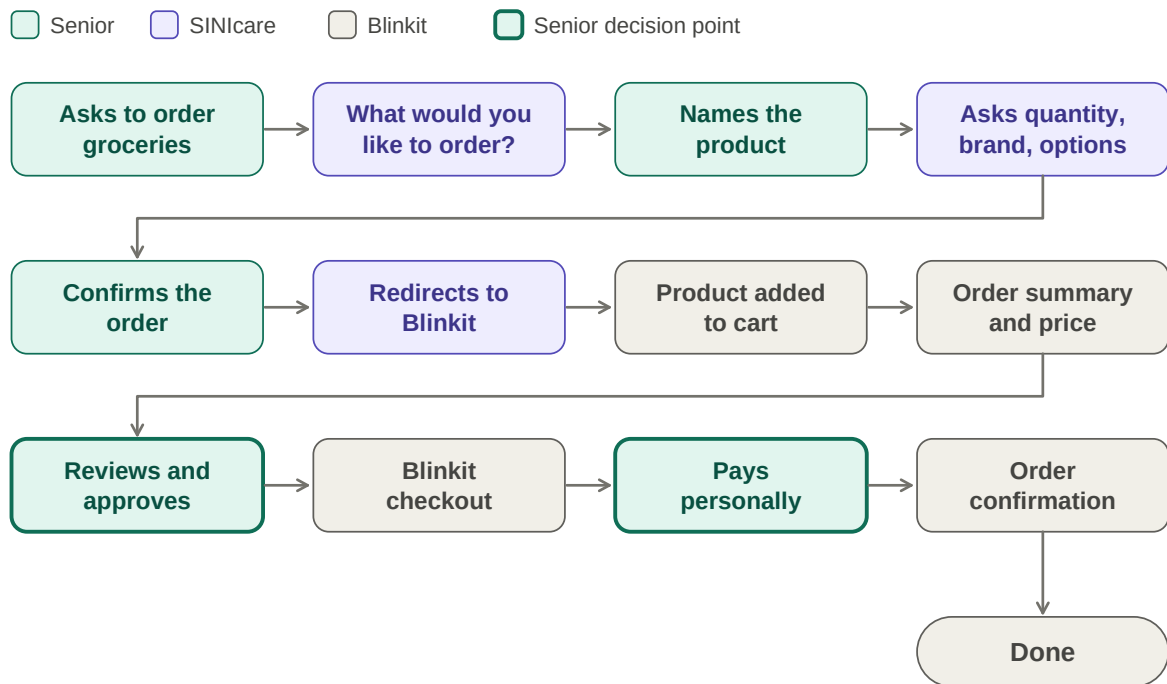
7.2 Help areas

The four everyday help areas. In each, SINicare does the confusing work and the senior stays in charge of the outcome.



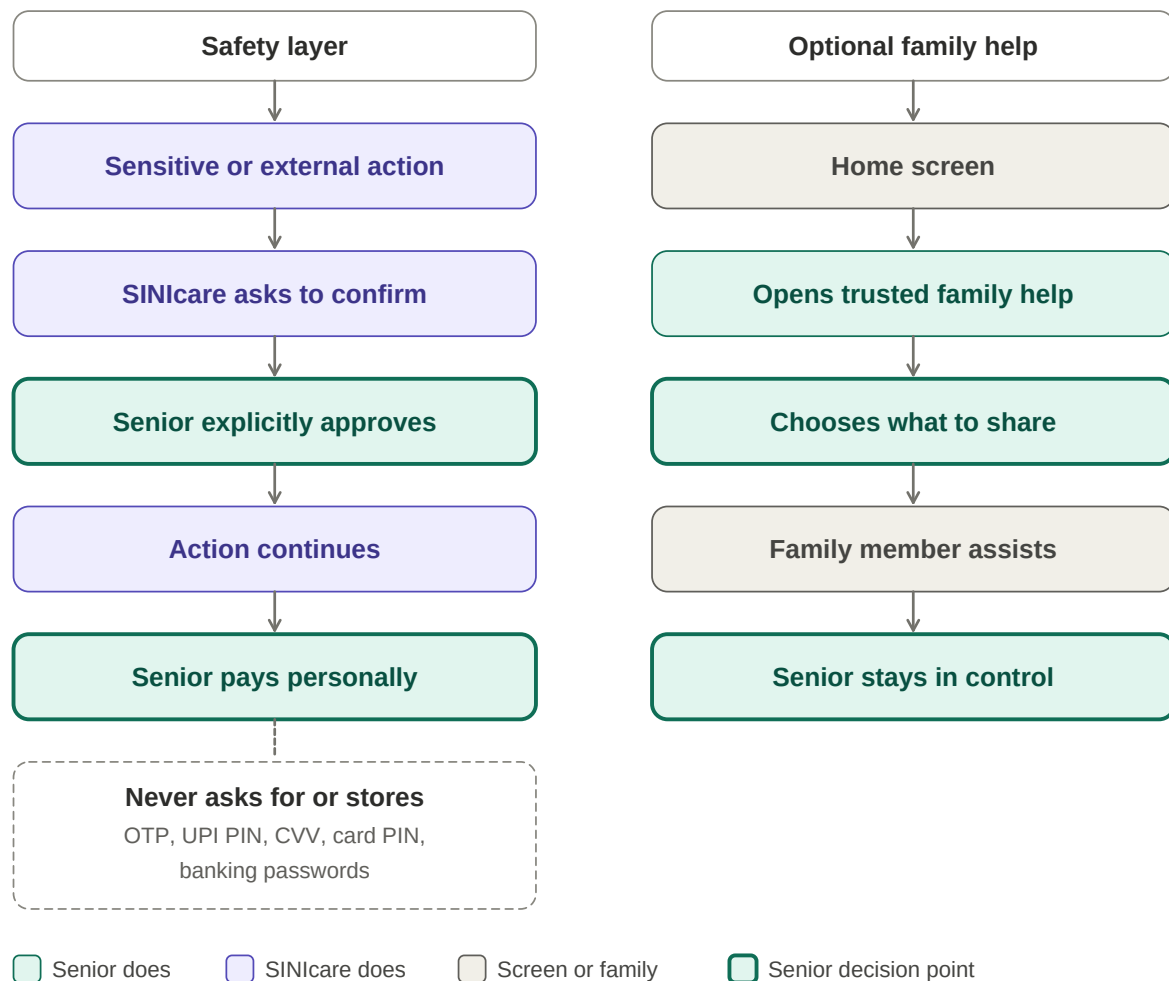
7.3 Grocery ordering (Blinkit hand-off)

SINicare gathers the order, then hands off to Blinkit. The senior reviews, approves, and pays personally; SINicare never completes a purchase on their behalf.



7.4 Safety layer and family help

Every sensitive or external action needs explicit approval. Family involvement is optional and limited to what the senior chooses to share.



Core principle: AI does the confusing work. The senior makes the important decisions.

8. UX and Accessibility Requirements

Visual design: calm, professional palette (deep blue/teal on warm white), high-contrast text, no decorative animation, respects prefers-reduced-motion.

- **Typography:** 20px minimum body text, 28px+ headings; user-adjustable text size (A / A+ / A++).
- **Touch targets:** at least 56 x 56px with generous spacing; icons always paired with text labels.
- **Navigation:** maximum four items (Ask, Check, Reminders, Family); persistent Home button.
- **Voice-first:** large mic button on every screen; confirmations spoken and shown.
- **Standards:** WCAG 2.2 AA minimum (contrast at least 4.5:1); full keyboard operation with visible focus states; ARIA labels and landmarks for screen readers; forgiving inputs (no timeouts that lose progress).
- **Language:** short sentences, non-technical words, no acronyms without explanation.
- **Responsive:** single-column layout that scales across mobile, tablet, and desktop.
- **Onboarding:** under 60 seconds: choose language, choose text size, done. No account needed to try; sign-in is required only to save reminders or add family.

9. AI Behaviour and Safety

- **Grounding:** system prompt enforces simple language, short answers, and a "tell me if unsure" rule; responses use structured output so the UI can render consistent cards.
- **Scam detection:** hybrid approach: rule-based signals (urgency, OTP/PIN requests, mismatched or shortened links, lookalike sender IDs, prize/KYC/"account blocked" patterns) combined with an LLM explanation. Rules give a floor: high-risk patterns are never rated "safe".
- **Guardrails:** no medical, legal, or investment advice (explain and suggest a professional); refuse to help share OTPs, passwords, or card details; escalate distress signals (for example, fear, coercion) with a suggestion to contact a trusted person or the national cyber-crime helpline (1930).
- **Transparency:** every answer is clearly AI-generated; "Was this helpful?" and "Ask a person" are always available.
- **Evaluation:** curated bilingual test set of scam and non-scam messages, document simplifications, and form flows, reviewed before each release.

10. Technical Approach

Architecture (modular and scalable)

- **Frontend:** React + TypeScript SPA/PWA, i18n layer (EN/HI), Web Speech API for STT/TTS with cloud fallback, component library built accessibility-first.
- **Backend:** thin API layer (Node or Python) with separate services/modules: companion, document, scam-check, reminders, family, auth. Each is behind an interface so new AI capabilities can be added without rewrites.
- **LLM gateway:** single abstraction over the model provider (prompt templates, structured outputs, rate limiting, caching, fallbacks).
- **Data:** PostgreSQL for users, reminders, and consent records; object storage for short-lived uploads.

Security and privacy

- Authentication via phone OTP or email link (no passwords for seniors); role-based access (senior vs trusted contact).
- TLS everywhere; encryption at rest; secrets in a vault; strict input validation and output encoding; file-type and size limits with malware scanning on uploads; rate limiting; CSRF/XSS/SSRF protections; dependency scanning.
- **Data minimisation:** uploads and message text deleted within 24 hours by default; no training on user data; PII redaction before model calls where possible; user can delete all data in one tap. Compliant with India's DPDP Act, with explicit consent for family sharing.

Efficiency: cache repeated queries and translations, stream responses, compress and resize images client-side before upload, lazy-load features, keep the initial bundle small for low-end devices and slow networks.

Testing: unit tests per module; integration tests for API and LLM gateway (with mocked model responses); contract tests for structured outputs; end-to-end tests for core journeys; automated accessibility checks (axe) plus manual screen-reader passes; the bilingual AI evaluation set runs in CI; usability testing with real seniors before launch.

11. Success Metrics

Area	Metric	MVP target
Activation	Users completing a first task in under 2 minutes	70%
Engagement	Weekly active users making at least 2 requests	40% of activated users
Task success	Tasks completed without abandonment or help	75%
Safety	Scam-check precision/recall on test set	Recall above 95% on high-risk messages
Trust	"Helpful" rating; self-reported confidence (pre/post survey)	85%+ helpful; measurable confidence lift

Area	Metric	MVP target
Accessibility	WCAG 2.2 AA audit; usability sessions with 10+ seniors	Zero critical issues
Reach	Share of sessions in Hindi	Tracked (informs language roadmap)

12. Risks and Mitigations

Risk	Mitigation
AI hallucination or wrong scam verdict	Rules-based floor, confidence language, "ask family" escape hatch, evaluation set gate
Over-reliance on AI	Encourages verification with official sources or family; never presented as authority
Voice recognition errors (accents, Hinglish)	Show transcript for confirmation before acting; easy retype
Privacy concerns with sensitive documents	Short retention, redaction, clear consent, one-tap delete
Senior can't get started	No-account trial, 60-second onboarding, family-assisted setup

13. Release Plan

Phase	Scope	Outcome
Alpha (weeks 1-6)	F1, F2, F3, F8 core; design system; accessibility baseline	Internal and family testing
Beta (weeks 7-12)	F4, F5, F7, F9; auth; security review	30-50 seniors, iterate on usability and scam accuracy
MVP launch (weeks 13-16)	F6; polish; monitoring; help/support flow	Public release
Post-MVP	More Indian languages, WhatsApp/SMS share-in, medication reminders, integrations with utilities and government services	Based on usage data

14. Open Questions

1. Sign-in method that works best for seniors: phone OTP vs. family-assisted setup?
2. Which SMS/reminder provider and cost model for reminders?
3. How much can be processed on-device vs. cloud to limit data exposure?
4. Do we need a human escalation path (helpline) for launch?